

• Homework 3b

Consider a spacecraft equipped with an attitude control system composed of 4 control momentum gyroscopes (single-gimbal, constant rotor rate) in a pyramid mounting, with angle $\beta = 54.74^\circ$ (angle between faces and base).

Inertia matrix of the entire spacecraft $J_c^{(0)} = \begin{bmatrix} 1200 & 100 & -200 \\ 100 & 2200 & 300 \\ -200 & 300 & 3100 \end{bmatrix} \text{ Kg m}^2$

Principal inertia moments of the rotors

$$I_s^{(j)} = 50 \text{ Kg m}^2 \quad I_r^{(j)} = 25 \text{ Kg m}^2 \quad (j=1, \dots, 4)$$

Rotor speed

$$\omega_r^{(j)} = \pi \text{ sec}^{-1} \quad (\text{i.e. } 30 \text{ rpm}) \quad \forall j$$

Initial angles for rotors

$$\theta_1^{(1)}(0) = 0 \quad \theta_1^{(2)}(0) = \frac{\pi}{4} \quad \theta_1^{(3)}(0) = \frac{\pi}{2} \quad \theta_1^{(4)}(0) = \frac{3\pi}{4}$$

Initial attitude

$$\underline{q}(0) = [0.5 \quad -0.6 \quad 0.2]^T \quad q_0 = \sqrt{1 - [q_1^2(0) + q_2^2(0) + q_3^2(0)]}$$

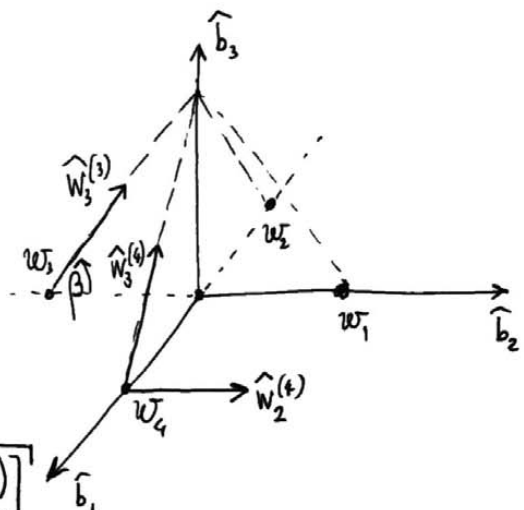
Initial angular rate

$$\underline{\omega}(0) = [0.1 \quad 0.05 \quad -0.05]^T \text{ sec}^{-1}$$

Final conditions

$$\underline{\omega}_f = [0 \quad 0 \quad 0]^T \text{ sec}^{-1}$$

$$\underline{q}_f = [0 \quad 0 \quad 0]^T, \quad q_{0,f} = 1$$



Geometry of mounting

$(\hat{w}_1, \hat{w}_2, \hat{w}_3)$ for each wheel

$$\hat{w}_i^{(j)} \equiv \hat{g}_j$$

$\hat{w}_3$ directed toward the vertex of the pyramid

Only a few among $\hat{w}_1^{(j)}, \hat{w}_2^{(j)}, \hat{w}_3^{(j)}$ are portrayed

For the gain matrices A and B , use the 2nd-order approximation with the following values of ζ and ω_n

$$\zeta = 1 \quad \omega_n = 0.05 \text{ sec}^{-1} \longrightarrow A^{-1} = c_1 I_{3 \times 3} \quad B = c_2 I_{3 \times 3}$$

These are the objectives, split in part (A) and part (B)

(A) Neglect the actuator dynamics; obtain the nonlinear control law (with compensation of the gyroscopic term) and apply it in numerical integration of the attitude dynamics, in the interval $0 \leq t \leq 300 \text{ sec}$. Plotting

- (i) time histories of $\underline{q}(t)$ and $q_0(t)$
- (ii) time histories of $\underline{\omega}(t)$
- (iii) time histories of $\underline{T}_c(t)$ (commanded torque components)

(B) Consider the actuator dynamics, i.e. the pyramid array of SG-CMGs, together with the nonlinear control law (with compensation of the gyroscopic term). Integrate the attitude dynamics in $0 \leq t \leq 300 \text{ sec}$. Plotting

- (i) time histories of $\underline{q}(t)$ and $q_0(t)$
- (ii) time histories of $\underline{\omega}(t)$
- (iii) time histories of $\underline{T}_c(t)$ (commanded torque components)
- (iv) time histories of $\underline{T}_a(t)$ (actual torque components)
- (v) time histories of $\underline{\dot{\theta}}(t)$ (i.e. $\dot{\theta}_1^{(1)}, \dots, \dot{\theta}_1^{(4)}$)